FORECASTING / PRODUCTION MLOPS

From a forecasting notebook to repeatable planning inputs

Make forecasts available through an API, automate repeatable workflows and keep model changes controlled.

WHY MOVE BEYOND THE NOTEBOOK?

Reliable delivery

Predictions for planning applications, without manual notebook runs and file exports.

Less manual work

Repeatable checks, training and releases with traceable results.

Controlled updates

Monitor quality, evaluate candidates and verify the rollback path.

Versioned releases

Model, store state and calendar artifacts belong to one release.

1 of 3 promoted

Three retraining events; one model promotion.

Rollback verified

100% traffic restored; readiness HTTP 200.

REPEATABLE FORECASTING, CONTROLLED MODEL UPDATES

Automate the path from data preparation to verified API serving. Evaluate candidates before replacing the active model.

01 Prepare & train

Repeatable feature and training steps.

02 Check candidate

Evaluate against the active model.

03 Bundle & verify

Version model and feature artifacts.

If promotion checks fail, the current Champion remains active.

Version	Registered at	Created by	Tags	Aliases
Version 11	10/01/2026, 10:16:51 AM		Add	@ champion
Version 10	10/01/2026, 10:12:18 AM		Add	Add
Version 9	10/01/2026, 10:10:06 AM		Add	Add

MLflow evidence: registered model versions and version 11 marked with the Champion alias. The controlled simulation recorded three retraining events and one promotion.

TESTED RECOVERY AFTER AN APPLICATION UPDATE

Cloud Run rollback

- Environment: `dev`
- Service: `mlops-sales-forecasting-dev-api`
- Revision: `mlops-sales-forecasting-dev-api-00001-brz`
- Traffic: `100%`
- Readiness: `HTTP 200`

FastAPI, Prefect, MLflow, XGBoost, Docker, GitHub Actions, Terraform, Cloud Run and Grafana.

Independent reference project using public Rossmann data. Promotion counts refer to a controlled simulation, not a client deployment. Cloud deployment and rollback successfully verified.

Have a working model? Start with a Notebook-to-MLOps Assessment.